Long-term low-protein, low-calorie diet and endurance exercise modulate metabolic factors associated with cancer risk.

BACKGROUND: Western diets, obesity, and sedentary lifestyles are associated with increased cancer risk. The mechanisms responsible for this increased risk, however, are not clear. OBJECTIVE: We hypothesized that long-term low protein, low calorie intake and endurance exercise are associated with low concentrations of plasma growth factors and hormones that are linked to an increased risk of cancer. DESIGN: Plasma growth factors and hormones were evaluated in 21 sedentary subjects, who had been eating a low-protein, low-calorie diet for 4.4 +/- 2.8 y (x +/- SD age: 53.0 +/- 11 y); 21 endurance runners matched by body mass index (BMI; in kg/m2); and 21 age- and sex-matched sedentary subjects eating Western diets. RESULTS: BMI was lower in the low-protein, low-calorie diet (21.3 +/- 3.1) and runner (21.6 +/- 1.6) groups than in the Western diet (26.5 +/- 2.7; P < 0.005) group. Plasma concentrations of insulin, free sex hormones, leptin, and C-reactive protein were lower and sex hormone-binding globulin was higher in the low-protein, low-calorie diet and runner groups than in the sedentary Western diet group (all P < 0.05). Plasma insulin-like growth factor I (IGF-I) and the concentration ratio of IGF-I to IGF binding protein 3 were lower in the low-protein, low-calorie diet group (139 +/- 37 ng/mL and 0.033 +/- 0.01, respectively) than in the runner (177 +/- 37 ng/mL and 0.044 +/- 0.01, respectively) and sedentary Western (201 +/- 42 ng/mL and 0.046 +/- 0.01, respectively) diet groups (P < 0.005). CONCLUSIONS: Exercise training, decreased adiposity, and long-term consumption of a low-protein, low-calorie diet are associated with low plasma growth factors and hormones that are linked to an increased risk of cancer. Low protein intake may have additional protective effects because it is associated with a decrease in circulating IGF-I independent of body fat mass.